

COMPREHENSIVE INVENTORY: WHAT'S IN GABER ET AL. PAPER VS. COMPLETE STATIC MALWARE ANALYSIS FIELD

Complete Coverage Audit (Section-by-Section + Technique-by-Technique)

PART A: WHAT THE PAPER COVERS (In Detail)

✓ SECTION 1: INTRODUCTION & MOTIVATION

WHAT'S THERE:

- ✓ Malware definition (trojans, ransomware, crypto miners, worms, botnets)
- ✓ Scale statistics (450K new samples/day, 970M samples by 2022)
- ✓ Cost of breaches (USD 4.35M average)
- ✓ Cyber arms race concept (evasion → mitigation → evasion)
- ✓ Anti-analysis techniques definition
- ✓ Threat model illustration (Figure 1)
- ✓ Problem statement (building accurate, robust AI models)
- ✓ 5 critical aspects framework

WHAT'S MISSING:

- ✗ Evolution narrative (how did field change 2015→2025?)
 - ✗ Paradigm shifts (classical ML → DL → representation learning)
 - ✗ Historical context (what was available in 2015 vs. now?)
 - ✗ Discipline maturity assessment (solved problems vs. open challenges)
 - ✗ Reference to EMBER2024 or modern benchmarks
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✓ SECTION 2: RELATED WORK (Prior Surveys Comparison)

WHAT'S THERE:

- ✓ 6 prior surveys compared (Gibert, Shaukat, Or-Meir, Aslan/Samet, Caviglione, Ucci)
- ✓ Table 1: Coverage matrix (shows their advantages)
- ✓ Gap identification (why prior surveys were incomplete)

- ✓ Explicit claim: "Most complete survey"
- ✓ Methodology transparency (what they'll cover)

WHAT'S MISSING:

- ✗ Post-2022 surveys (SOREL-20M era papers, 2023+ surveys)
 - ✗ Quantitative comparison (metrics on breadth, depth, recency)
 - ✗ Weakness analysis of prior surveys (why were they outdated?)
 - ✗ Citation of this paper's own limitations upfront
 - ✗ Explicit temporal scope statement ("2015-2025 survey")
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✓ SECTION 3: METHODOLOGY (Search Strategy & Criteria)

WHAT'S THERE:

- ✓ Research questions (RQ1-RQ6)
- ✓ Multiple databases (ACM DL, IEEE Xplore, Scopus, Google Scholar)
- ✓ Explicit search strings (reproducible)
- ✓ Inclusion criteria (80% peer-reviewed minimum)
- ✓ Exclusion criteria (non-English, no explicit AI technique, missing results)
- ✓ Final count (77 papers selected)
- ✓ Grey literature allowed (20% max)
- ✓ Temporal range (2018-2022 focus)

WHAT'S MISSING:

- ✗ LLM-assisted screening methodology
 - ✗ Inter-rater reliability for inclusion/exclusion
 - ✗ Citation chasing methodology (papers citing EMBER?)
 - ✗ Author contact for unpublished/preprint papers
 - ✗ Sensitivity analysis (what if different databases used?)
 - ✗ Quality scoring of included papers (systematic bias assessment)
 - ✗ Data extraction template
 - ✗ Systematic review registration (PROSPERO, OSF)
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✓ SECTION 4.1: MALWARE SOPHISTICATION (Evasion Techniques)

4.1.1 Evasive Malware

WHAT'S THERE:

- ✓ **Obfuscation techniques:**
 - Code packing (compression, DLL unhooking)
 - Metamorphic (dead code insertion, CPU register reassignment, code reordering)
 - Polymorphic (encryption, code mutation, decryption on-demand)

- Commercial packers (UPX, Themida, etc.)

- **✓ Anti-Analysis techniques (Detailed):**

- Anti-debugging (IsDebuggerPresent, CheckRemoteDebuggerPresent, OutputDebugString, etc.)
- Anti-VM (registry checks, CPU features, CPUID)
- Anti-DBI (code cache artifacts, overhead detection)
- Anti-sandboxing (timing, environment fingerprinting)
- Table 2: 4 operation types (IP, API, syscalls, memory access)
- Table 3: 17 evasive techniques with prevalence data

- **✓ Empirical prevalence:**

- Galloro et al. [30]: 92 techniques, 80% malware are evasive
- Park et al. [74]: 29 techniques, 8 detect Intel Pin
- Kim et al. [51]: 16.21% anti-DBI prevalence
- Polino et al. [77]: 15.6% anti-DBI prevalence
- Sharma et al. [83]: 99.36% anti-analysis prevalence
- Maffia et al. [59]: 68% use obfuscation
- Branco et al. [15]: 88.96% use ≥ 1 anti-analysis

- **✓ Evasion circumvention:**

- DBI framework hooks, bypass mechanisms, blacklist of terms
- Intel Pin plugins for anti-evasion

WHAT'S MISSING:

- **✗ Control-flow obfuscation:**

- Control-flow flattening (CFG flatness increases)
- Bogus jumps / opaque predicates
- Indirect jumps (JMP [RAX+RBX])
- Virtual machine-based obfuscation (VM exit/entry overhead)

- **✗ Data-flow obfuscation:**

- Variable renaming
- Constant unfolding
- Data structure transformation

- **✗ Semantic obfuscation:**

- Code transformations preserving semantics but defeating syntactic analysis
- Type erasure
- Constant propagation

- **✗ Packer-specific techniques:**

- Self-modifying code detection/evasion
- IAT (Import Address Table) hooks
- TLOC (Thread Local Storage) callbacks

- **✗ Anti-analysis arms race evolution:**
 - How obfuscation changed 2015→2025
 - Malware response to detection improvements
 - **✗ Unpacking methods:**
 - Generic unpacking (heuristic-based)
 - PolyUnpack [47] approach
 - Emulation-based unpacking
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4.1.2 Novel Malware

WHAT'S THERE:

- ✓ Definition: Malware without signature, not fitting families, zero-day exploits
- ✓ APT (Advanced Persistent Threat) characteristics
- ✓ APT examples:
 - SolarWinds attack (HAFNIUM, supply-chain)
 - Stuxnet (SCADA systems, zero-day exploits)
- ✓ Zero-day exploitation (4 zero-days in Exchange Server)
- ✓ Custom exploits and TTPs
- ✓ Accessibility of malicious tools (open-source, frameworks)

WHAT'S MISSING:

- **✗ Malware family evolution:**
 - How families are defined (behavioral similarity vs. genetic similarity?)
 - Family tree reconstruction
 - Phylogenetic analysis of malware
- **✗ Zero-day malware detection:**
 - How to detect without signatures?
 - Behavioral heuristics for zero-day detection
 - Metamorphic zero-days
- **✗ Supply-chain malware:**
 - Provenance tracking
 - Bill-of-materials (BOM) analysis
 - Transitive trust chains
- **✗ Ransomware-specific evolution:**
 - Ransomware-as-a-Service (RaaS) infrastructure
 - Decryption key management
 - Payment tracking

- **✗ Worm propagation:**
 - Network-based spread mechanisms
 - Exploit kit evolution
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4.1.3 AI-Powered Malware

WHAT'S THERE:

- **✓ DeepLocker (IBM BlackHat 2018)**
 - CNN target detection via webcam
 - Encrypted payload decryption on target match
 - Black-box neural network as encryption key
- **✓ GUI-based attacks (Yu et al.)**
 - TensorFlow Object Detection for browser icons
 - Stealth login emulation
 - Potential banking attack extension
- **✓ AI-powered targeting & trigger conditions**
- **✓ Black-box NN for payload encryption**
- **✓ Concept: NNs hide malicious intent**

WHAT'S MISSING:

- **✗ AI-powered evasion:**
 - GAN-based feature perturbation
 - Adversarial example generation
 - Evasion while preserving functionality
 - **✗ Reinforcement learning malware:**
 - RL agents learning evasion policies
 - Game-theoretic adversary-detector dynamics
 - **✗ LLM-generated malware:**
 - ChatGPT/Copilot malware generation capability
 - Polymorphic code generation
 - **✗ Malware with learned defenses:**
 - Detector-aware evasion
 - Adaptive adversarial malware
 - **✗ Supply-chain attack automation:**
 - Automated persistence mechanisms
 - Lateral movement strategies
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✓ SECTION 4.2: ANALYSIS TECHNIQUES

4.2.1 Static Analysis

WHAT'S THERE:

- ✓ **PE file structure (Figure 2):**
 - Header, Sections, imports, memory mapping, execution code
- ✓ **Static analysis tools (Table 4):**
 - IDA Pro: Disassembly, opcodes, header, functions, strings, CFG
 - Pefile (Python): PE header, sections, strings, imports
 - Peframe (Python): API calls, DLLs
 - Binary file bytes: N-grams analysis
 - VirusTotal: Labeling
- ✓ **Features extracted:**
 - Opcode sequences, frequencies
 - Byte sequences, n-grams
 - PE file headers, sections, sizes
 - String artifacts
 - Control flow graphs (implicit)
- ✓ **Limitations acknowledged:**
 - Obfuscation (68% prevalence defeats static)
 - Polymorphism, metamorphism
 - Packing limitations
- ✓ **Novel approaches:**
 - PE-to-image conversion (RGB/grayscale)
 - Byte patterns as pixels
 - Image classification on malware images
- ✓ **Limitation of image conversion:**
 - Not effective on novel malware
 - Defeated by benign carrier apps

WHAT'S MISSING:

- ✗ **Advanced static analysis:**
 - **Symbolic execution:** Constraint solving, path exploration
 - **Taint analysis:** Data flow from sources (file reads) to sinks (network sends)
 - **Type inference:** Reconstructing types from binaries

- **Program slicing:** Identifying relevant code portions
 - **✗ Intermediate Representation (IR) analysis:**
 - Binary lifting (Valgrin, BAP, LLVM IR)
 - Semantic-invariant features
 - Decompilation quality (Hex-Rays, Retargetable Decompiler)
 - **✗ Control Flow Graph (CFG) analysis:**
 - CFG extraction algorithms (recursive descent, linear sweep, hybrid)
 - CFG canonicalization (handling jumps, indirect calls)
 - CFG comparison metrics (graph isomorphism, subgraph matching)
 - **✗ Data Flow Graph (DFG) analysis:**
 - DFG extraction from IR
 - Data dependency identification
 - Reaching definitions analysis
 - **✗ Code similarity detection:**
 - Graph edit distance
 - Subgraph isomorphism
 - Semantic similarity (beyond syntactic)
 - **✗ Disassembly correctness:**
 - Ambiguous bytes (can be data or code)
 - Indirect jump targets
 - Function boundary detection errors
 - **✗ Cross-architecture analysis:**
 - x86 vs. x86-64 differences
 - ARM, MIPS, PowerPC handling
 - Architecture-agnostic features
 - **✗ Malware-specific static techniques:**
 - API import table analysis (IAT hooking detection)
 - Section permission analysis (.text writable?)
 - Anomalous header fields
 - Suspicious string detection (URLs, registry keys)
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4.2.2 Sandbox Analysis (Dynamic Analysis)

WHAT'S THERE:

- **✓ Sandbox definition:** Controlled environment for behavior observation

- **✓ Advantages over static:**
 - Resolves obfuscation (execution decrypts code)
 - Captures runtime behavior
 - Not defeated by syntactic obfuscation
- **✓ Cuckoo Sandbox (Table 5):**
 - API calls (Windows API monitoring)
 - Registry access
 - DLL loading
 - File operations
 - Network activity (DNS, HTTP)
 - Process creation
 - Memory operations
 - Configuration: Timeout, analysis options
- **✓ Features extracted:**
 - API call sequences
 - Registry keys accessed
 - Files modified/deleted
 - Network traffic (IPs, domains, protocols)
 - Dropped files
- **✓ Anti-sandbox techniques:**
 - Trigger mechanisms (click, user input required)
 - Function call branching (different behavior if unhooked)
 - Environment checks (number of processes, disk size)
- **✓ Limitations:**
 - Time-dependent behavior (triggers after N days)
 - User interaction required (clicks)
 - Stealthiness (detects if running in sandbox)

WHAT'S MISSING:

- **✗ Advanced dynamic analysis:**
 - **DRAKVUF:** Kernel-level monitoring via hypervisor introspection
 - **SecBox:** Container-based sandboxing on Linux
 - **Bitvisor:** Lightweight VMM for dynamic analysis
 - **Frida:** Runtime hooking framework
- **✗ Memory forensics:**
 - Memory dump analysis (heap, stack)
 - Injected code detection
 - Code cave hunting
 - Stolen API recovery
- **✗ Hardware performance counters:**
 - Cache behavior

- Branch prediction patterns
 - Virtualization overhead detection
 - **✗ Network traffic analysis:**
 - Encrypted traffic decryption (MITM proxy)
 - C2 (Command & Control) communication patterns
 - Botnet callback detection
 - DNS sinkhole detection evasion
 - **✗ Real machine analysis:**
 - Behavior on production systems
 - Persistence mechanisms
 - Lateral movement attempts
 - **✗ Temporal analysis:**
 - How behavior changes over time
 - Delayed payload execution
 - Time-bombed malware
 - **✗ Multi-stage execution:**
 - Dropper → Stage 2 → Stage 3 chains
 - Injection chains (process injection, DLL injection)
 - **✗ Evasion-aware dynamic analysis:**
 - Anti-sandbox countermeasures (fake process trees, spoofed times)
 - DRAKVUF transparency improvements
 - Zero-copy analysis (leave execution unmodified)
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4.2.3 Dynamic Binary Instrumentation (DBI)

WHAT'S THERE:

- **✓ Intel Pin framework:**
 - Instruction-level monitoring
 - Runtime hooking
 - Plugin-based extensions
 - Overhead: 1-5x slowdown
- **✓ Evasion circumvention (Table 3):**
 - DBI-detection evasion techniques (8 techniques can detect Pin)
 - Bypass mechanisms implementation
 - Blacklist-based environment spoofing
 - Time-reduced reporting (for timing checks)

- **✓ Detection of anti-DBI:**

- QueryInformationProcess (QIP) hooks
- RDTSC (CPU timestamp) monitoring
- Unhandled exception filter
- GetTickCount/QueryPerformanceCounter

- **✓ Plugin development for anti-evasion:**

- Prefix handling
- Memory breakpoint detection
- Self-modification handling

WHAT'S MISSING:

- **✗ Other DBI frameworks:**

- **DynamoRIO:** Open-source, efficient
- **QEMU:** Emulation-based, high overhead
- **Valgrind:** Instrumentation framework

- **✗ DBI transparency challenges:**

- NtQuerySystemInformation (SYSTEMPROCESSINFORMATION) changes
- Processor affinity detection
- Instruction cache artifacts
- JIT compilation transparency

- **✗ DBI performance optimization:**

- Code caching
- Fragment linking
- Coarse-grain instrumentation

- **✗ Advanced DBI features:**

- Conditional instrumentation (instrument only suspicious operations)
- Shadow execution (parallel clean/instrumented execution)
- Hybrid static-dynamic (CFG extraction via DBI)

- **✗ Kernel-level instrumentation:**

- Syscall interception
 - Kernel API hooking
 - Hypervisor-based monitoring (DRAKVUF)
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✓ SECTION 4.3: MALWARE REPOSITORIES (Datasets)

WHAT'S THERE:

- ✓ **EMBER (2018):**
 - 1.1M samples
 - Binary classification (malware/benign)
 - Pre-extracted static features
 - No binaries (copyright, safety)
- ✓ **BODMAS (2019-2020):**
 - 134K samples
 - 14 malware categories
 - Timestamp-aware (enables temporal analysis)
- ✓ **SOREL-20M (2017-2019):**
 - 20M samples (large-scale)
 - Time-split recommendations
 - Addresses EMBER saturation
- ✓ **VirusShare:**
 - 55M+ samples
 - Unlabeled (requires VirusTotal consensus)
 - Live/ongoing collection
 - Benign software: Legitimate applications
- ✓ **Malware Bazaar:**
 - 700K+ samples
 - Structured taxonomy (family, category)
 - Recent samples
 - API rate limits
- ✓ **Limitations noted:**
 - Dataset imbalance (real 10:90, benchmark 50:50)
 - Concept drift (2015 malware ≠ 2024 malware)
 - Labeling quality (VirusTotal consensus can be wrong)
 - Class imbalance between families

WHAT'S MISSING:

- ✗ **EMBER2024 (Released Jun 2025):**
 - 3.2M samples (vs. EMBER 1.1M)
 - 6 file formats (PE, ELF, Mach-O, DEX, APK, etc.)
 - 7 tasks (not binary classification)
 - Evasion-focused challenge set

- Published baselines
 - **✗ Dataset comparisons:**
 - Feature compatibility (EMBER features $\neq$ SOREL features?)
 - Temporal overlap (which datasets from which years?)
 - Cross-dataset performance (model trained on EMBER, tested on VirusShare)
 - **✗ Synthetic datasets:**
 - GAN-generated malware (address imbalance)
 - Adversarial example datasets
 - **✗ Behavioral datasets:**
 - **CIC-MalMem-2022:** Memory snapshots, obfuscated malware
 - **Drebin:** Android APKs (dated 2010-2012)
 - **CIC-IDS:** Network-based intrusion detection
 - **UNSW-NB15:** Network traffic
 - **✗ Real-world datasets:**
 - Private vendor datasets (Microsoft, McAfee)
 - Enterprise environment samples
 - Supply-chain attack samples
 - **✗ Dataset versioning & evolution:**
 - How EMBER changed over time
 - Why SOREL-20M emerged (EMBER saturation?)
 - Successor datasets to SOREL
 - **✗ Labeling methodology:**
 - How VirusTotal consensus calculated?
 - Confidence scores for labels
 - False positive rates in benchmarks
 - **✗ Feature extraction tools:**
 - EMBER feature extractor version changes
 - Reproducibility across tool versions
 - Feature correctness (bugs in PE parsing)
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✓ SECTION 4.4: FEATURE SELECTION

WHAT'S THERE:

- **✓ Feature categories:**
 - Static: Opcodes, PE headers, strings, n-grams
 - Dynamic: API calls, registry, network, syscalls

- Hybrid: Both static + dynamic
- **✓ Feature engineering importance:**
 - Quality features determine model quality
 - Manual feature engineering labor-intensive
 - Domain expertise required
- **✓ Feature dimensionality:**
 - PE header (79 static features in AI-Hydra [103])
 - API sequences (513 features in AI-Hydra [103])
 - Combined high-dimensional
- **✓ Feature selection challenges:**
 - Irrelevant features add noise
 - Correlated features redundancy
 - Benign software uses same APIs (false positives)

WHAT'S MISSING:

- **✗ Feature selection methods:**
 - Correlation-based: Remove correlated features
 - Mutual information: Information-theoretic importance
 - Chi-square test: Categorical feature importance
 - Recursive feature elimination (RFE)
 - L1 regularization (LASSO automatic selection)
- **✗ Dimensionality reduction:**
 - PCA (Principal Component Analysis)
 - t-SNE / UMAP (visualization)
 - Autoencoders (learned representations)
- **✗ Feature engineering techniques:**
 - N-gram extraction (1-grams, 2-grams, 3-grams)
 - Byte pair encoding (BPE)
 - Hashing (to reduce dimensions)
- **✗ Learned representations:**
 - Deep learning automatically learns features
 - Convolutional filters learn patterns
 - Embedding layers (semantic spaces)
 - Self-supervised pre-training (no manual features)
- **✗ Domain-specific features:**
 - Packing indicators (entropy, section names)
 - Suspicious APIs (CreateRemoteThread, WriteProcessMemory)
 - Network behavior (DNS queries, HTTP requests)
 - Ransomware indicators (file encryption, payment sites)

- **✗ Adversarially robust features:**

- Which features are robust to obfuscation?
 - Which features can adversary manipulate?
 - Interpretable ML feature importance
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✓ **SECTION 4.5: MACHINE LEARNING VS. DEEP LEARNING**

WHAT'S THERE:

- **✓ Classical ML techniques:**

- Random Forest (RF)
- Support Vector Machines (SVM)
- Logistic Regression (LR)
- Naive Bayes (NB)
- Decision Trees
- **Results:** Competitive with DL on some benchmarks

- **✓ Deep Learning techniques:**

- Dense Neural Networks (DNN)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Deep Belief Networks (DBN)
- **Advantages:** Better on complex patterns

- **✓ Figure 4 & Table 8:**

- Comparison of techniques
- Accuracy metrics across papers
- Model complexity, training time

- **✓ Hybrid approaches:**

- Combining static + dynamic features
- Multiple model ensemble

- **✓ Generative AI / LLM mention:**

- ChatGPT, BARD briefly mentioned
- 4 papers on LLM + cybersecurity
- Not integrated into methodology

WHAT'S MISSING:

- **✗ Advanced ML techniques:**

- **Gradient Boosting:** XGBoost, LightGBM (often beat DL on EMBER)
- **Isolation Forests:** Anomaly detection

- **One-class SVM:** Out-of-distribution detection
- **✗ Deep Learning architectures:**
 - **Attention mechanisms:** Transformer variants
 - **Graph Neural Networks (GNN):** GCN, GAT, GIN on CFG
 - **Variational Autoencoders (VAE):** Generative models
 - **Generative Adversarial Networks (GAN):** Adversarial examples
- **✗ Representation learning:**
 - **Self-supervised pre-training:** Pre-train on unlabeled data
 - **Contrastive learning:** Siamese networks, triplet loss
 - **Embeddings:** Word2vec, DeepWalk for malware representations
- **✗ Transfer learning:**
 - Pre-training on large datasets → fine-tune
 - Domain adaptation (EMBER → VirusShare)
 - Few-shot learning (novel families with <10 samples)
- **✗ Ensemble methods:**
 - Voting (majority, weighted)
 - Stacking (meta-learner)
 - Cascade (fast filter → slow deep analysis)
- **✗ Interpretable ML:**
 - Decision trees (glass-box)
 - Rule-based systems
 - SHAP, LIME, Grad-CAM for explanations
- **✗ AutoML & hyperparameter optimization:**
 - Grid search, random search, Bayesian optimization
 - Neural architecture search (NAS)
- **✗ LLM integration:**
 - **Full section missing** on LLM-assisted malware analysis
 - CodeLLaMA, GPT-4 for assembly understanding
 - Semantic lifting (decompilation)
 - Few-shot prompting

✓ SECTION 5: DISCUSSION (Trends & Challenges)

WHAT'S THERE:

- ✓ Challenges identified:
 - Obfuscation defeats static analysis
 - Anti-analysis defeats dynamic analysis
 - Novel malware requires generalization

- Dataset imbalance
- Concept drift
- ✓ Deployment reality:
 - Need for practical, lightweight models
 - Computational constraints
 - False positive implications

WHAT'S MISSING:

- ✗ Systematic open problems list
- ✗ Prioritization of challenges (impact × feasibility)
- ✗ Future research roadmap
- ✗ Post-2025 outlook
- ✗ Unresolved fundamental questions

PART B: COMPREHENSIVE FIELD COVERAGE MATRIX

Static Malware Analysis Complete Field Breakdown

``` STATIC ANALYSIS SUBFIELD COVERAGE IN PAPER COMPLETENESS

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DISASSEMBLY & LIFTING: ─ Disassembly algorithms ~2 pages 15% ─ IDA Pro / Ghidra tools ~1 page 10% ─ Binary lifting (IR) 0 pages 0% **✗✗✗** CRITICAL GAP ─ Decompilation 0 pages 0% **✗✗✗** CRITICAL GAP ─ Cross-architecture 0 pages 0% **✗✗✗** CRITICAL GAP ─ Semantic recovery 0 pages 0% **✗✗✗** CRITICAL GAP

OBFUSCATION & EVASION: ─ Packing/compression ~2 pages 40% ─ Polymorphism ~2 pages 40% ─ Metamorphism ~2 pages 40% ─ Control-flow flattening 0 pages 0% **✗✗** MAJOR GAP ─ Data-flow obfuscation 0 pages 0% **✗✗** MAJOR GAP ─ Semantic obfuscation 0 pages 0% **✗✗** MAJOR GAP ─ Anti-disassembly 0.5 pages 10% ─ Unpacking methods 0 pages 0% **✗** GAP

FEATURE EXTRACTION: ─ Opcode n-grams ~3 pages 50% ─ Byte sequences ~2 pages 40% ─ PE headers ~2 pages 50% ─ API imports ~2 pages 40% ─ String analysis ~1 page 30% ─ Control flow graphs ~1 page 20% ─ Data flow graphs 0 pages 0% **✗** GAP ─ Symbolic execution 0 pages 0% **✗** GAP ─ Taint analysis 0 pages 0% **✗** GAP ─ Program slicing 0 pages 0% **✗** GAP

SIMILARITY & CLASSIFICATION: ─ Graph edit distance 0 pages 0% **✗** GAP ─ Subgraph matching 0 pages 0% **✗** GAP ─ Semantic similarity 0 pages 0% **✗** GAP ─ Family classification ~2 pages (excluded) N/A ─ Clustering methods 0 pages 0% **✗** GAP

LEARNING MODELS: ─ Classical ML (RF, SVM) ~3 pages 60% ─ CNN (Convolutional) ~2 pages 40% ─ RNN (Recurrent) ~2 pages 40% ─ GNN (Graph Neural Networks) ~0.1 pages 5% **✗✗✗** CRITICAL GAP ─



Transformers ~0.5 pages 5% **xxx** CRITICAL GAP | LLM-assisted ~0.3 pages 5% **xxx** CRITICAL GAP | Few-shot / Meta-learning 0 pages 0% **xx** MAJOR GAP | Representation learning 0 pages 0% **xx** MAJOR GAP | Ensemble methods ~1 page 20%

EVALUATION & ROBUSTNESS: | Temporal evaluation ~0.5 pages 5% **xx** MAJOR GAP | Cross-dataset testing ~0.5 pages 5% **xx** MAJOR GAP | Adversarial robustness ~1 page 10% **xx** MAJOR GAP | GAN-based evasion ~1 page 15% **xx** MAJOR GAP | Certified defenses 0 pages 0% **xx** MAJOR GAP | False positive rates ~0.5 pages 10% | Explainability (XAI) 0 pages 0% **xx** MAJOR GAP

DATASETS & BENCHMARKS: | EMBER ~2 pages 50% | SOREL-20M ~1 page 30% | BODMAS ~0.5 pages 20% | VirusShare ~0.5 pages 15% | EMBER2024 0 pages 0% **xxx** CRITICAL GAP | Dataset comparison ~1 page 20% | Class imbalance ~0.5 pages 15% | Concept drift ~0.5 pages 5%

TOOLS & REPRODUCIBILITY: | IDA Pro / Ghidra ~2 pages 40% | Cuckoo Sandbox ~2 pages 50% | Intel Pin DBI ~2 pages 50% | Binary lifting frameworks 0 pages 0% **x** GAP | Tool versioning 0 pages 0% **x** GAP | Containerization 0 pages 0% **x** GAP | Reproducibility protocols 0 pages 0% **xx** MAJOR GAP

PRACTICAL DEPLOYMENT: | Scalability (450K/day) 0.5 pages 5% **xx** MAJOR GAP | Inference latency 0.5 pages 5% **xx** MAJOR GAP | Resource requirements 0 pages 0% **xx** MAJOR GAP | Tiered pipelines 0 pages 0% **xx** MAJOR GAP | Production architectures 0 pages 0% **xx** MAJOR GAP | Cost-benefit analysis 0 pages 0% **xx** MAJOR GAP

OPEN PROBLEMS & FUTURE: | Semantic obfuscation resistance 0 pages 0% **xx** MAJOR GAP | Cross-temporal generalization 0 pages 0% **xx** MAJOR GAP | Real-time detection 0 pages 0% **x** GAP | LLM robustness 0 pages 0% **x** GAP | Research roadmap ~0.5 pages 5% **x** GAP

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OVERALL FIELD COVERAGE: ~32% CRITICAL GAPS (0% coverage): 15 topics MAJOR GAPS (5-10% coverage): 20 topics ````

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## PART C: DETAILED TOPIC-BY-TOPIC BREAKDOWN

### 1. DISASSEMBLY & BINARY LIFTING

**Paper Coverage:** ~2 pages (IDA Pro + Pefile mentioned) **Missing:**

- **x** Disassembly correctness (ambiguous bytes, indirect jumps)
- **x** Linear sweep vs. recursive descent algorithms
- **x** Ghidra analysis

- **✗ Binary lifting frameworks (Valgrin, BAP, LLVM IR) ✗✗** CRITICAL
- **✗ Decompilation quality (Hex-Rays, Retargetable Decompiler) ✗✗** CRITICAL
- **✗** Semantic bugs in lifters [web:20]
- **✗** Cross-architecture lifting (x86 → ARM → MIPS)

## 2. OBFUSCATION TECHNIQUES

**Paper Coverage:** ~6 pages (packing, polymorphism, metamorphism, anti-analysis) **Missing:**

- **✗ Control-flow flattening ✗✗** MAJOR
- **✗ Data-flow obfuscation ✗✗** MAJOR
- **✗ Semantic obfuscation ✗✗** CRITICAL
- **✗** Indirect jumps / opaque predicates
- **✗** Virtual machine-based obfuscation
- **✗** Self-modifying code evasion
- **✗** Unpacking techniques (generic, emulation-based)
- **✗** Commercial packer analysis (UPX, Themida, ASPack)

## 3. FEATURE EXTRACTION

**Paper Coverage:** ~3 pages (opcodes, PE headers, n-grams) **Missing:**

- **✗** Feature selection methods (correlation, mutual information, RFE)
- **✗ Dimensionality reduction (PCA, t-SNE, autoencoders) ✗✗** MAJOR
- **✗ Symbolic execution for feature generation ✗** GAP
- **✗ Taint analysis features ✗** GAP
- **✗ Program slicing ✗** GAP
- **✗** Domain-specific features (packing indicators, suspicious APIs)
- **✗** Adversarially robust feature extraction
- **✗** Feature importance analysis

## 4. GRAPH ANALYSIS (CFG/DFG)

**Paper Coverage:** ~1 page (CFG mentioned briefly) **Missing:**

- **✗ Graph Neural Networks (GCN, GAT, GIN) ✗✗** CRITICAL
- **✗ CFG extraction algorithms ✗✗** MAJOR
- **✗ CFG canonicalization ✗✗** MAJOR
- **✗ Data Flow Graph (DFG) extraction ✗** GAP
- **✗ Graph edit distance ✗** GAP
- **✗ Subgraph isomorphism ✗** GAP
- **✗ Graph embeddings (node2vec, DeepWalk) ✗** GAP
- **✗** CFG-based similarity detection
- **✗** Graph robustness to control-flow flattening

## 5. DEEP LEARNING ARCHITECTURES

**Paper Coverage:** ~4 pages (CNN, RNN, DNN) **Missing:**

- **× Graph Neural Networks ××× CRITICAL**
- **× Transformers (BinBert, Vision ViT) ××× CRITICAL**
- **× Attention mechanisms ×× MAJOR**
- **× Variational Autoencoders (VAE) × GAP**
- **× Generative Adversarial Networks (GAN) × GAP**
- **× LSTM vs. GRU considerations × GAP**
- **× Hyperparameter tuning strategies**
- **× Neural architecture search (NAS)**

## 6. REPRESENTATION LEARNING

**Paper Coverage:** 0 pages **Missing:**

- **× Self-supervised pre-training ××× CRITICAL**
- **× Contrastive learning ××× CRITICAL**
- **× Embeddings (word2vec, DeepWalk) ××× CRITICAL**
- **× Transfer learning paradigm ×× MAJOR**
- **× Domain adaptation ×× MAJOR**
- **× Few-shot learning ×× MAJOR**
- **× Meta-learning (MAML, Prototypical Networks) ×× MAJOR**
- **× Metric learning × GAP**

## 7. LARGE LANGUAGE MODELS (LLM)

**Paper Coverage:** ~0.3 pages (4 papers mentioned) **Missing:**

- **× CodeLLaMA for assembly understanding ××× CRITICAL**
- **× GPT-4/Gemini binary analysis ××× CRITICAL**
- **× Semantic lifting via LLM ××× CRITICAL**
- **× Few-shot learning with LLMs ×× MAJOR**
- **× LLM robustness to adversarial inputs ×× MAJOR**
- **× Prompt engineering for malware analysis × GAP**
- **× Chain-of-thought prompting × GAP**
- **× Edge deployment of LLMs**

## 8. EXPLAINABILITY & INTERPRETABILITY

**Paper Coverage:** 0 pages **Missing:**

- **× SHAP (SHapley Additive exPlanations) ×× MAJOR**
- **× LIME (Local Interpretable Model-Agnostic) ×× MAJOR**
- **× Grad-CAM / Gradient-based attribution ×× MAJOR**
- **× Integrated Gradients × GAP**
- **× Attention weight visualization × GAP**
- **× Counterfactual explanations × GAP**
- **× Feature importance (permutation-based) × GAP**
- **× Why interpretability matters for security**

## 9. ADVERSARIAL ROBUSTNESS

**Paper Coverage:** ~1 page (anti-analysis techniques) **Missing:**

- **✗ GAN-based adversarial malware ✗✗ MAJOR**
- **✗ Adversarial example generation (FGSM, PGD) ✗✗ MAJOR**
- **✗ Certified defenses (IBP, randomized smoothing) ✗✗ MAJOR**
- **✗ Threat model definition ✗ GAP**
- **✗ Adversarial training ✗ GAP**
- **✗ Game-theoretic analysis ✗ GAP**
- **✗ Detector-aware malware adaptation ✗ GAP**
- **✗ Robustness evaluation frameworks ✗ GAP**

## 10. EVALUATION FRAMEWORKS

**Paper Coverage:** ~2 pages (mentions challenges) **Missing:**

- **✗ Temporal evaluation methodology ✗✗ MAJOR**
- **✗ Concept drift quantification ✗✗ MAJOR**
- **✗ Cross-dataset evaluation protocol ✗✗ MAJOR**
- **✗ Time-stratified train/test splits ✗ GAP**
- **✗ FPR at production-relevant thresholds ✗ GAP**
- **✗ Precision-recall curves (vs. ROC/AUC) ✗ GAP**
- **✗ Cost matrix evaluation ✗ GAP**
- **✗ Reproducibility standards ✗ GAP**

## 11. DATASETS & BENCHMARKING

**Paper Coverage:** ~4 pages (EMBER, SOREL, BODMAS, VirusShare)  
**Missing:**

- **✗ EMBER2024 ✗✗✗ CRITICAL**
- **✗ Dataset saturation issue (EMBER AUC 0.999) ✗ GAP**
- **✗ Feature extraction tool correctness ✗ GAP**
- **✗ Cross-dataset generalization ✗ GAP**
- **✗ Synthetic datasets (GAN-based malware) ✗ GAP**
- **✗ Private vendor datasets ✗ GAP**
- **✗ Real-world imbalanced evaluation ✗ GAP**
- **✗ Dataset versioning & evolution ✗ GAP**

## 12. PRACTICAL DEPLOYMENT

**Paper Coverage:** ~1 page (mentions latency) **Missing:**

- **✗ Scalability analysis (450K samples/day) ✗✗ MAJOR**
- **✗ Inference latency benchmarks ✗✗ MAJOR**
- **✗ Resource requirements (memory, GPU, CPU) ✗✗ MAJOR**
- **✗ Tiered detection pipeline design ✗✗ MAJOR**
- **✗ Production architectures ✗ GAP**
- **✗ Hardware accelerators ✗ GAP**
- **✗ Model compression (quantization, distillation) ✗ GAP**

- **✗ Edge deployment constraints ✗ GAP**

### 13. REPRODUCIBILITY

**Paper Coverage:** 0 pages **Missing:**

- **✗ Tool versioning impact ✗✗ MAJOR**
- **✗ Containerization (Docker) ✗✗ MAJOR**
- **✗ Feature extraction reproducibility ✗ GAP**
- **✗ Open-source vs. proprietary tools ✗ GAP**
- **✗ Code availability standards ✗ GAP**
- **✗ Data sharing governance ✗ GAP**
- **✗ Systematic review registration ✗ GAP**

### 14. OPEN PROBLEMS & FUTURE

**Paper Coverage:** ~0.5 pages (brief conclusion) **Missing:**

- **✗ Systematic problem enumeration ✗✗ MAJOR**
- **✗ Problem prioritization (impact ✗ feasibility) ✗✗ MAJOR**
- **✗ Unresolved fundamental questions ✗ GAP**
- **✗ Post-2025 research roadmap ✗ GAP**
- **✗ Established vs. speculative trends ✗ GAP**
- **✗ Emerging attack surfaces ✗ GAP**

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## PART D: QUANTITATIVE SUMMARY

### Coverage by Category (% of Complete Field)

Category Coverage Status

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Disassembly & Lifting 15% **✗✗✗ CRITICAL** Obfuscation & Evasion 50% **✗✗ NEEDS WORK** Feature Extraction 40% **✗✗ NEEDS WORK** Graph Analysis (CFG/DFG) 20% **✗✗✗ CRITICAL** Deep Learning Architectures 35% **✗✗✗ CRITICAL** Representation Learning 0% **✗✗✗ CRITICAL** Large Language Models 5% **✗✗✗ CRITICAL** Explainability & Interpretability 0% **✗✗ MAJOR GAP** Adversarial Robustness 15% **✗✗ MAJOR GAP** Evaluation Frameworks 25% **✗✗ MAJOR GAP** Datasets & Benchmarking 65% **✗ MINOR GAP** Practical Deployment 20% **✗✗ MAJOR GAP** Reproducibility 0% **✗✗ MAJOR GAP** Open Problems & Future 10% **✗ MINOR GAP**

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OVERALL FIELD COVERAGE ~27% INADEQUATE FOR 2025

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## PART E: WHAT THE PAPER DOES WELL (Strengths)

### ✓ WHAT'S EXCELLENT IN THE PAPER

#### 1. Evasion Techniques Taxonomy (~6 pages with detailed tables)

- ✓ Most comprehensive evasion technique enumeration
- ✓ Empirical prevalence data (92 techniques, 80% malware)
- ✓ Tables 2 & 3 with specific technique detection rates
- ✓ Circumvention strategies documented

#### 2. Dataset Comparison (~4 pages)

- ✓ EMBER, SOREL, BODMAS coverage
- ✓ Scale and characteristics noted
- ✓ Limitations discussed (imbalance, drift)

#### 3. Classical ML vs. DL (~4 pages)

- ✓ Balanced comparison
- ✓ Trade-offs explained
- ✓ Figure 4 visual comparison

#### 4. Analysis Techniques (~4 pages)

- ✓ Cuckoo Sandbox detailed (Table 5)
- ✓ DBI (Intel Pin) explained with anti-evasion
- ✓ Practical tools listed

#### 5. Organization & Structure

- ✓ 5-pillar framework logical
- ✓ Easy to navigate
- ✓ Clear research questions

#### 6. Methodology Transparency

- ✓ Explicit search strings
- ✓ Inclusion/exclusion criteria
- ✓ 77 papers systematically analyzed

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## PART F: CRITICAL GAPS FOR 2025 COMPETITIVENESS

### ● MUST-HAVE FOR 2025 CREDIBILITY (0% coverage)

| Topic               | Pages Missing | Impact              |
|---------------------|---------------|---------------------|
| Binary lifting & IR | 3-4           | Semantic robustness |

| Topic                  | Pages Missing      | Impact                            |
|------------------------|--------------------|-----------------------------------|
| Graph Neural Networks  | 4-5                | SOTA for CFG analysis             |
| Transformers           | 5-6                | SOTA architecture                 |
| LLM-assisted analysis  | 6-7                | Paradigm shift                    |
| EMBER2024 dataset      | 1-2                | Latest benchmark                  |
| Adversarial robustness | 3-4                | Detector vulnerability            |
| Explainability (XAI)   | 2-3                | SHAP/LIME/Grad-CAM                |
| Temporal evaluation    | 2-3                | Concept drift framework           |
| <b>TOTAL MISSING</b>   | <b>27-34 pages</b> | <b>Would bring to 60-67 pages</b> |

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## PART G: FINAL VERDICT

**Overall Field Coverage: ~27% (Should be 80%+ for tier-1 survey)**

**Paper Strengths:** Evasion taxonomy, dataset overview, classical analysis techniques

**Paper Weaknesses:** Modern paradigms (GNN, Transformers, LLM), robustness, deployment, evaluation frameworks

**Recommendation:**

- ✓ Good for 2020-2021 snapshot
  - ✗ Insufficient for 2025 survey standards
  - Needs 25-35 pages of new content to be competitive
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**END OF COMPREHENSIVE INVENTORY**